

# Milestone One Concept Memo

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**Assisted Wheelchair Transfer System**

**IED Section:** 5

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## Problem Statement

The design team was introduced to a kindergarten-aged boy with several physical disabilities in a local school district. Note that to protect anonymity, this boy will be referred to as “the user” or “the individual” and no personal details will be mentioned in this report. Through interviews with an educator of his, the team learned that the user has motor control issues, language and communication issues, and mobility issues arising from an ataxic gait pattern, also known as Ataxia. An Ataxic gait pattern refers to an unsteady, uncoordinated walk that is often characterized by a lack of balance and poor coordination [1]. Around 150,000 people in the US have some sort of Ataxia, some from birth, and some developed later in life [2]. Though the severity of symptoms varies, many of these individuals are dependent on wheelchairs and walkers. For these people, independence in mobility is a huge problem. Independent mobility can lead them to work more easily, need fewer caregivers, and have more confidence. The complexities of cerebellum disorders that lead to an ataxic gait pattern are far beyond the scope of this project, the team is instead focusing on this specific user’s needs. The individual’s issues currently confine him to a special education classroom. This classroom has a high teacher/assistant to student ratio which allows the user to get the attention that he needs for help with mobility, but due to the fact that most of the other classmates have mental disabilities he hasn’t been developing academically as much as his educators believe he has potential for. As the team understands, they plan for the user to join the main cohort of kids in the “standard” class pathway. There are challenges to this, as the user currently relies on his teachers’ and assistants’ help with a variety of day-to-day tasks. From holding a pencil to speaking to walking, there are many obstacles for the user. The design team is hoping to design a solution that will help the user overcome some of these obstacles so that he will have greater independence in his new classroom. The design team will primarily focus on his mobility issue, more specifically getting up out of his wheelchair and into his Crocodile Gait Trainer that he uses as a walking aid. This transition is currently done either with the help of a wall or the aid of an assistant. Ideally, our solution would allow the customer to do this themselves. Some of the expected benefits of the solution are:

- Increased independence and confidence for the user
- Allows for user to advance into standard classpath
- Less strain on the user
- Space efficiency in the classroom
- Lower need from caregivers

If this project is successful it could be used for other people with similar gait issues, or even people with other mobility issues such as the elderly. The design team of engineers feels that this project is important and meaningful for them. Even if the proposed solution does not solve every problem the user faces, it is a worthwhile and fulfilling project to be involved in as it directly impacts the well-being of an individual.

## Customer Requirements

As mentioned in the Problem Statement section of this Memo, the design team was introduced to a kindergartener with this specific neurological condition called Ataxia gait, which affects an individual's coordination, and those with the condition would typically stagger and lose their balance constantly [1]. The kindergartener wants a device that can transition its user from a sitting position to a standing position safely and seamlessly so that he can transition from special ed class to a regular k-12 elementary school. Other customers that the product is targeting would be someone who is either a late kindergartener or an early elementary school, born with an ataxic gait pattern or someone who has a similar disability with the same symptoms as the kindergartener, someone who was born with a disability that requires them to use a tool like a wheelchair or a walker to travel and cannot transition seamlessly from sitting to standing or vice versa to switch from a wheelchair to a walker. Other stakeholders for this product are teachers, caregivers, and fellow classmates of the customers, as intended benefits for the user carry over to them due to time saved in the classroom.

To accommodate the needs of our target customers, several requirements are mandatory to ensure that the final product fits the customer's needs. Given that the customer will most likely use the product daily for school, the product must not break after a couple of uses, and it should be able to last for several years before problems and errors occur. The product must work reliably each time it is used, it must be consistent with no variation in how long it takes for the transition nor have any unexpected problems occurring from it. Our product targets disabled youth with an age range of around 7 to 9, meaning that it has to be convenient, not taxing to use, and must only have a maximum of 4 moving parts combined to keep it simple and grounded. The customer is also probably looking for something easily portable and does not take up much space making it easy to store in compartments or cars for later usage. Finally, the product must not be too expensive for the parents of the targeted customer with the total cost somewhere between 150 to 350 dollars. The technical and performance specifications are estimated below in Table 1:

- The force that the product can handle per use ranges from 45 to 75 newtons because the customer will require something to grasp onto during the transition between standing and sitting as a way to balance themselves. These values are estimated by taking into account the average mass of a 7 to 9-year-old, using it to calculate the force of gravity acting on them, and then taking a rough guess of 20 percent to 50 percent to find the force they can exert when pushing against something[12].

- The time before small problems or errors appear ranges from 2.5 to 3 years and is estimated to have a rough guess of how long the product is going to last using an assessment of the design team's skills and abilities. This specification is added because being consistent and knowing when the product will start to deteriorate will positively impact the safety of the customer.
- The number of moving parts in the product should not exceed a limit of 4 because any more would make it harder to control and keep track of, a product with 5 or more moving parts would seem inconvenient and not at all simple like what the customer requires.
- The maximum weight the device can shift without the wheelchair moving ranges from 22 to 29 kg, this range is the upper limit at which the product will not cause the wheelchair to move when the transition between sitting to standing or vice versa is occurring. These values are averages for both males and females ages 7 to 9 [11].
- The height that the device can lift someone to ranges from 120 to 134 cm, these values are generalized by using data on average male and female height between the ages of 7 and 9 years old [11].
- The maximum weight of the product is around 8 to 12 kg because that is the average weight for transport wheelchairs and taking into account the age of the targeted customers [13], the targeted range is estimated.
- The max height of the device is estimated from the typical average height of a walker being around half of the height of the customer [14], therefore the range of the maximum height of the product ranges from 60 to 67 centimeters, where the customer heights used to calculate this range came from the average of height for male and female between the ages of 7 and 9 years old [11].

*Table 1: Customer Requirements and Technical Specifications*

| <b>Customer Requirement</b>                                        | <b>Technical/Performance Specification</b>                        | <b>Target Value / Range of Values</b> |
|--------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------|
| Durability                                                         | Force the product can handle per use                              | 45-95 N                               |
| Consistency                                                        | Time before small problems or errors appears                      | 2.5-3 years                           |
| Convenient, simple, easy to operate, and will not exhaust the user | Maximum number of moving parts.                                   | 4                                     |
|                                                                    | Age of the user                                                   | 7-9 years                             |
| Stability                                                          | Maximum weight the device can shift without the wheelchair moving | 22-29kg                               |
|                                                                    | Height that the device can lift someone to                        | 120-134 cm                            |
| Inexpensive from the parent's perspective                          | Dollars per person                                                | 150-350\$                             |

| Customer Requirement | Technical/Performance Specification | Target Value / Range of Values |
|----------------------|-------------------------------------|--------------------------------|
| Portable             | Maximum Weight of product           | 8-12 kg                        |
|                      | Max height                          | 60-67 cm                       |

## Concepts and Benchmarking

After preliminary research and interviews with people in charge of patients' everyday mobility, developing a table of competitive products is a challenge. The majority of products that assist in a patient's mobility are either too big and bulky or have no autonomous function integrated into the design. Ideally, our concept allows the user full autonomy so they do not have to wait on a nurse or aid to help them to just go to the bathroom or walk across the room. A bulk of products in the current market focus on making the life of the caregiver easier in the form of easily moveable carts and electronic lifts that are not controlled or activated by the patient. For example the Rifton Tram, a fully customizable mobility device with adjustable wheelbase and height and a scale still relies on the nurse or caregiver whom the patient needs to rely on to move around [17]. Other products like the Seveni Patient Lift Aid are cheaply made of unenforced aluminum brackets with step lock breaking mechanisms that can only be activated using precise foot movements [18]. Most other products don't provide the necessary stability to suit a customer with cerebral palsy if they do not have the correct build quality and hand breaking. From our research, there are no reliable, stable, and cost-efficient options for most patients afflicted by cerebral palsy to move around independently. By creating a wheelchair attachment that can be used on different styles and sizes of wheelchair we can provide an independent experience for the user. Most of these metrics in the table below are hard to define quantitatively and, as such, they are not. However, stability in a wheelchair is an obvious goal, but it cannot be defined as simply as the wheelbase. This is a reliable metric used in four-wheeled modes of transportation, but other factors come into account when determining overall stability. One such factor is the size of the wheel. In general, a device with six wheels is less susceptible to tipping forces than a four-wheeled device. The diameter of the wheels also becomes a concern when regarding tipping in the left and right directions. Table 2, below, references a series of existing products used for personal mobility whose product descriptions claim that they are applicable to cerebral palsy. Most of them have highly scored reviews, but the main problem we aim to tackle is gaining a more independent mobility experience for the user.

Table 2: Benchmarking and existing products(item sources are found in citation 14-18)

| Metric      | Unit                 | Orthomen<br>Electric<br>Transfer | Seveni<br>Patient Lift<br>Aid | Elenker<br>HKF-9405 | Rifton Tram   |
|-------------|----------------------|----------------------------------|-------------------------------|---------------------|---------------|
| Cost        | \$                   | 1800 (sale)                      | 320                           | 475                 | unlisted      |
| Stability   | in.<br>(wheelbase)   | 22                               | 23                            | 16                  | 22-55         |
| Portability | in. (when<br>folded) | does not fold                    | does not fold                 | does not fold       | does not fold |
| Autonomy    | N/A                  | x                                | x                             | x                   | x             |

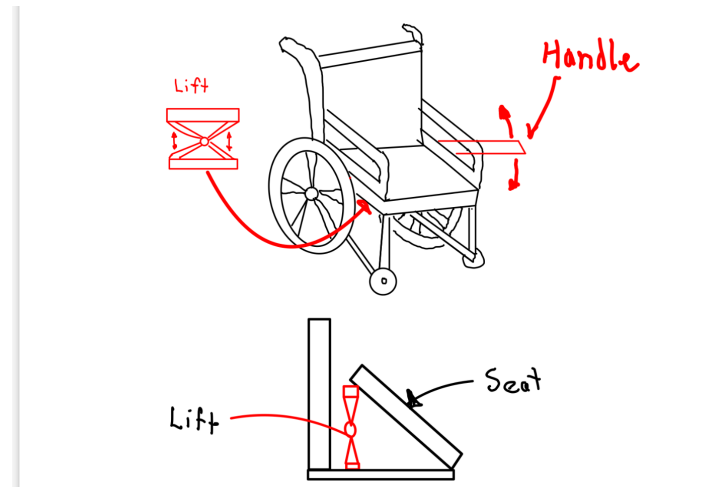


Figure 1: Concept one, scissor lift

The team's first concept involved a rising seat that would push the user up via a scissor lift type of mechanism (shown in Figure 1). This, coupled with a fold-out handle for stability through the whole process would be ideal for supporting the user through the whole process. Being autonomous, however, brings safety risks if it fails and does not allow for the user to utilize his upper body strength and would cause the user to lose their balance and fall. There could be many moving parts which all have the potential to break and not work reliably, these moving parts also present a challenge to the simplicity of the product and could be hard to control.

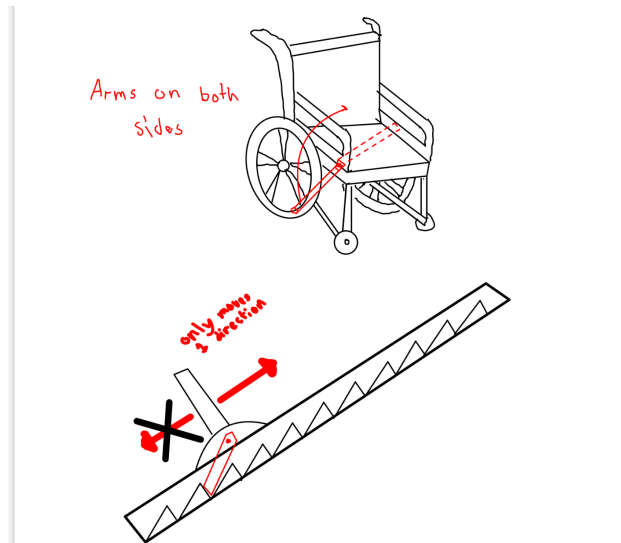


Figure 2: Concept two, ratchet rails

As the team learned more about the user, concepts were honed in on his strengths. A concept using ratchets on rails that the user would pull on to pull themselves up (shown in Figure 2). Utilizing the user's upper body strength and keeping them on a straight track up. Overall this concept seems promising with its mechanism and could work going up or down, but the user is unsupported once they let go of the handle which causes problems such as the user following over or falling face down. It would also need an anti-tipping device as an attachment to keep the wheelchair from rolling over during the incline, attachments like this already exist [3] and could be attached to a lower bar on the wheelchair as a remedy to that problem.

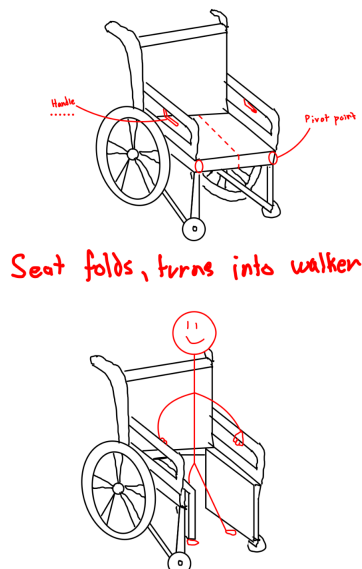
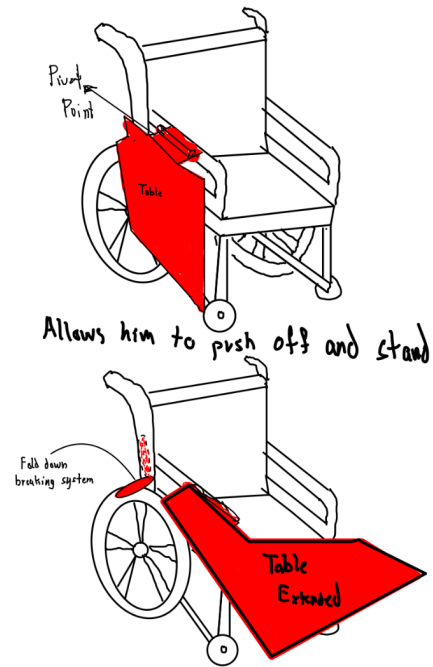


Figure 3: Concept three, Seat drop

Concept three is a significantly different idea. It involves drastically changing the wheelchair so that there exists a mechanism where the seat folds down so it could be used as a walker (shown

in Figure 3). However, this concept would involve a lot of alterations to the wheelchair itself which deviates from the original intention of making an attachment to the wheelchair and could potentially be unsafe as there isn't really any support for when the seat drops out. It also may take away from the usefulness of a gait trainer, as the weight of the wheelchair would be a drag on the user making the product exhausting and tedious to use.



*Figure 4: Concept four, fold-out table*

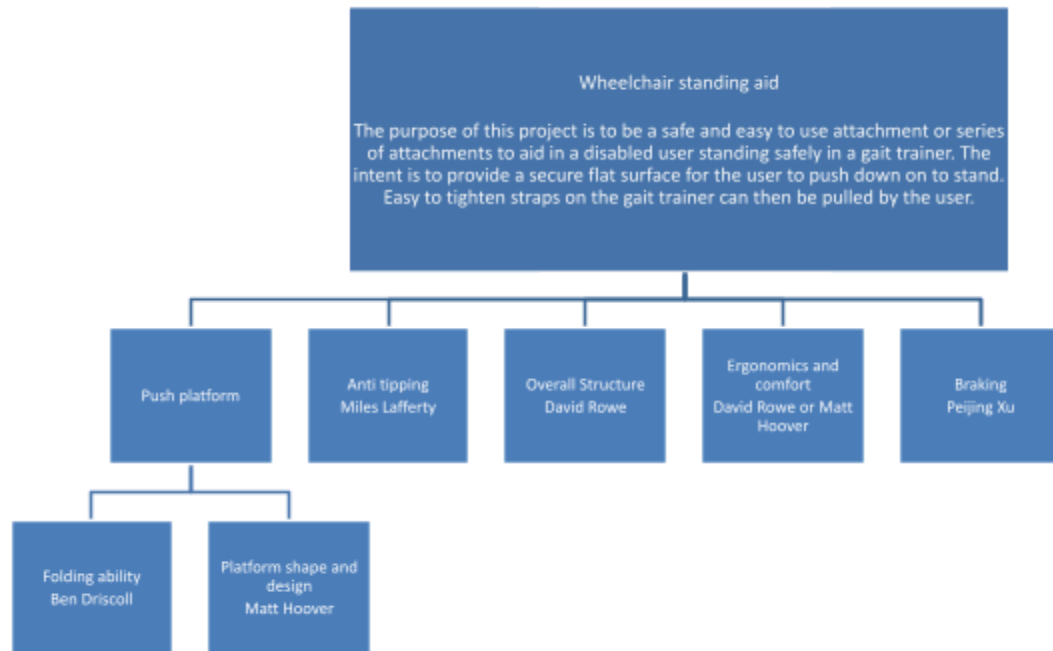
When the design team was made aware of the fact that the user can already use a table to stand up, concept four was developed. This allows the customer to use the standing-up method they already use wherever and whenever. It brings the table to the customer when they want to transition from sitting to standing. Made of either one or two fold-out tables that have a space for the user's body once they stand up (shown in Figure 4). There again is a risk of tipping forward when the user puts his whole weight on the table and the transition from the wheelchair and walker also needs to be fleshed out. Some additions to this concept would be straps to lock his legs in place or other stabilizing system that prevents the user from losing their balance.

The group assembled a concept selection matrix (Table 3) with all the concepts listed. For criteria, the group compiled customer requirements and added team instinct. For each criterion +1 means that the concept meets the criteria, 0 means the concept does not have a clear impact on the selection criteria, and -1 means the concept negatively impacts the criteria. For team instinct, the team voted on each concept. The points are collected in the net score row. Concept 4 had by far the highest tally. The team chose to combine the anti-tipping mechanism from concept 2 to add to concept four as it had the same issues of eccentric loading when the user pushes down on the tables.

Table 3: Concept Selection Matrix

|                            | Concepts     |               |                         |                                                     |
|----------------------------|--------------|---------------|-------------------------|-----------------------------------------------------|
|                            | 1            | 2             | 3                       | 4                                                   |
| Selection Criteria         | Scissor lift | Ratchet rails | Wheelchair/walker combo | Folding tables                                      |
| Size minimized             | 0            | 0             | 1                       | 0                                                   |
| Weight minimized           | 0            | 0             | -1                      | 1                                                   |
| Affordability              | -1           | -1            | 0                       | 1                                                   |
| Robustness                 | 0            | 1             | 1                       | 1                                                   |
| Stable through procedure   | -1           | -1            | 1                       | -1                                                  |
| Utilizes user's upper body |              | 1             | -1                      | 1                                                   |
| Constructability           | 1            | 1             | 1                       | 1                                                   |
| Team Instinct              |              | 1             | -1                      | 1                                                   |
| Moving parts               | -1           | -1            | 1                       | 1                                                   |
| Sum of +1's                | 1            | 4             | 5                       | 7                                                   |
| Sum of 0's                 | 0            | 0             | 0                       | 0                                                   |
| Sum of -1's                | -3           | -3            | -3                      | -1                                                  |
| Net Score                  | -2           | 1             | 2                       | 6                                                   |
| Rank                       | 4            | 3             | 2                       | 1                                                   |
| Continue?                  | No           | No            | No                      | Yes - combine with concept 2 anti-tipping mechanism |

## Proposed Solution



*Figure 5: Subsystem Hierarchy Diagram*

Our proposed solution is a lightweight and efficient attachable design. After getting an accurate description of how our customer currently stands and gets into his gate trainer, we thought a simple, built-in, foldable table design would be the most efficient and effective way for a student to stand up easily. The subsystems include the overall structure, linking other subsystems to the existing wheelchair and Walker. With our braking system, we cannot have either the wheelchair or the Walker moving when our customer is standing or in the process of standing. Our stabilization/anti-tipping system will help with the necessity for the user's body to be supported during the sanding process. In our ergonomics and comfort system, our user must be able to return to the seated position after standing practically and comfortably. Our Push Platform system could be adjusted depending on how much we change this concept. We have two subsystems for our push platform: folding ability and platform shape and design. Our design will have a hinging point attached to the side of the chair, which holds the table in a vertical plane relative to the chair so it stays out of the way and doesn't take up too much space. When the user wants to stand, he can move the table at its pivot point, which is attached to the chair, and slide the table in front of the customer so it is suspended in the air in front of the wheelchair. The customer can then use his standard technique for standing with this table while the brakes on this wheelchair are locked into place. Once standing, he can migrate from the table to his Walker. From there, whenever he'd like to sit, he reverses the process using the table to sit back down into the chair, folding the table away, and it's back to normal in the wheelchair. As you can see in

the figure below, I've made a subsystem hierarchy diagram that expresses the importance of each subsystem and emphasizes our process of making this project successful.

## Project Plan

Table 4: Group Gantt Chart

| Project Name                     | Status | Owner      | Start    | End      |
|----------------------------------|--------|------------|----------|----------|
| <b>Milestone 1 MEMO</b>          | 100%   |            | 10/20/24 | 10/22/24 |
| Concept Design 3 + models        | 100%   | Miles L    | 10/20/24 | 10/22/24 |
| Problem Statement                | 100%   | Ben D      | 10/20/24 | 10/22/24 |
| Customer requirements            | 100%   | Peijing Xu | 10/20/24 | 10/22/24 |
| Technical specifications         | 100%   | Peijing Xu | 10/20/24 | 10/22/24 |
| Benchmarking                     | 100%   | Matt H     | 10/20/24 | 10/22/24 |
| Concept selection table (scored) | 100%   | Ben D      | 10/20/24 | 10/22/24 |
| Chosen solution                  | 100%   | David R    | 10/20/24 | 10/22/24 |
| Project plan                     | 100%   | Ben D      | 10/20/24 | 10/22/24 |
| Materials                        | 100%   | David R    | 10/20/24 | 10/22/24 |
| Editing                          | 50%    | Miles L    | 10/22/24 | 10/23/24 |
| Submit MEMO                      | 0%     | Matt H     | 10/23/24 | 10/23/24 |
| Review concepts with Jill        | 0%     | Matt H     | 10/24/24 | 10/31/24 |
| Purchase Materials               | 0%     | David R    | 10/24/24 | 11/7/24  |
| Prototype subsystem 1            | 0%     | Ben D      | 10/24/24 | 11/10/24 |
| Prototype subsystem 2            | 0%     | David R    | 10/24/24 | 11/10/24 |
| Prototype subsystem 3            | 0%     | Matt H     | 10/24/24 | 11/10/24 |
| Prototype subsystem 4            | 0%     | Miles L    | 10/24/24 | 11/10/24 |
| Prototype subsystem 5            | 0%     | Peijing Xu | 10/24/24 | 11/10/24 |
| <b>Testing</b>                   | 0%     |            |          |          |
| Subsystem integration test       | 0%     | Peijing Xu | 11/7/24  | 11/24/24 |
| Test on real wheelchair          | 0%     | David R    | 11/25/24 | 12/1/24  |
| Test with user                   | 0%     | Matt H     | 12/1/24  | 12/8/24  |
| <b>Milestone 2 DEMO</b>          | 0%     | Ben D      | 12/3/24  | 12/3/24  |
| Milestone 3 Presentation         | 0%     |            |          |          |
| Milestone 3 MEMO                 | 0%     |            |          |          |

This project will be one centered on iterations of prototyping and testing. Tasks are kept track of by a group Gantt chart (shown in Table 3). The Gantt chart is updated continuously. The design team plans to present this proposed solution to the contact (the user's educator) and receive feedback. After that, the prototype of each subsystem will be developed and tested. Each subsystem is led by an individual group member but it is expected that all members will provide support for all parts of the project. Materials will be purchased and costs shared by all members of the group. The testing plan is still to be formally developed.

## Materials

The following materials are the essential components required to create the wheelchair attachment. Each component was thought about carefully, changes will be made if there are better options that are more cost-effective. These components ensure high-quality products that are both functional and safe. The wheelchair attachment focuses on comfort and ease of use. We have also included a brief description of the parts and functions below.

Table 5: Preliminary Parts List

| Need/Function           | Part Type          | Part Name                           | Price        | Retailer        |
|-------------------------|--------------------|-------------------------------------|--------------|-----------------|
| Wheelchair Stability    | Anti-Tip Mechanism | Karman Anti-Tippers [3]             | \$40         | Home Depot      |
| Braking Function        | Braking Mechanism  | Quickie Push to Lock Wheel Lock [4] | \$90         | DME Hub         |
| Help standing up        | Table              | Wood Desktop [5]                    | \$70         | Etsy            |
| Body Stability          | Handles            | Push Handles [6]                    | \$50         | EZ Lite Cruiser |
| Protection from falling | Cushions           | Wheelchair Cushions [8]             | \$40         | Vcomsoft        |
| Movement for table      | Hinge              | Surface Mount Hinge [7]             | \$20         | Home Depot      |
| <b>TOTAL</b>            |                    |                                     | <b>\$320</b> |                 |

- **Anti-Tip Mechanism:** Included to prevent the wheelchair from tipping forward when the user begins to stand or lean forward and to prevent the wheelchair from falling.
- **Braking Function:** The wheel lock will be used so when standing up the wheelchair will be set in place to prevent it from rolling and will be complemented with the anti-tippers so it won't flip.
- **Help standing up:** The Wood Desktop is a wooden slab that is going to be used as a surface for the user to push down onto to lift themselves up into the walker.
- **Body Stability:** Push handles to hold onto so that the user can hold it to maneuver if need be to the table to push off or to stand up.
- **Protection from falling:** The wheelchair cushions are in place in case the user loses balance and falls back into the seat to protect from any possible injuries from the fall.
- **Movement for the table:** These hinges are required to adjust the wood desktop positioning to allow it to be hidden when not in use and stable when in use.

## **Professional Development**

This project's design team is made up of members with a diverse set of technical and soft skills. Two aerospace engineers, two computer systems engineers, and a civil/structural engineer. This allows the group to come up with a wide variety of solutions.

In terms of traits no two members have the exact same MBTI "personality types". However, certain traits do occur most frequently in the group. INTJ "architect" is the team's most frequent, corresponding with "creative ingenuity, straightforward rationality, and self-improvement" [2]. The team plans on utilizing these strengths throughout the design process. Some traits are less common on the team, and these must be acknowledged and worked on for the team to be successful. The lacking ESFP "entertainer" personality type corresponds with strong interpersonal skills. Openly disagreeing with something can be tough for introverts and that could mean that the team needs to work on sharing feelings. Presentations and customer interviews are also likely to be uncomfortable for the group so practice and rehearsals will be scheduled in. In terms of conflict management, the group is fairly spread out, with some being more collaborative, accommodating, compromising, or competing. Communicating different ideas and opinions will be vital to conflict resolution so the team has agreed to "team standards agreements" that lay out methods for resolving conflicts.

## **Conclusion**

The design team seeks to design and construct a product of several mechanical wheelchair attachments to make the transfer process for an individual from a wheelchair to a gait trainer more streamlined and independent. This project should allow the user more independence and thus greater academic potential. The current solution (subject to change) consists of two folding chairs that would allow the user to push against to let the user up and down, coupled with a number of other subsystems. The design team has a plan in place to design, test, evaluate, and iterate all aspects of the solution to arrive at a working product.

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<https://www.amazon.com/Transfer-Wheelchair-Disabled-Bedridden-toileting/dp/B0C77YJTWB>